

Training Robot Swarms for Adaptive Foraging in Environments with Obstacles

Pigar Biteng¹, Tameem Uz Zaman¹, and Qi Lu²

Abstract—We apply the NeuroEvolution of Augmented Topologies (NEAT) to train adaptive and efficient swarm robotic foraging behavior in unknown environments with random obstacles. By rewarding effective actions and penalizing inefficient ones, the system promotes coordination and obstacle avoidance, minimizing redundant exploration and outperforming traditional stochastic foraging algorithms. The optimization focuses on cumulative reward fitness, evaluated through simulations with three types of distributed resources, where swarm performance is analyzed in terms of foraging time efficiency and resource retrieval rates. Comparative experiments conducted across two different swarm sizes demonstrate that the elaborated penalty-reward strategy enhances resource retrieval rates while reducing energy expenditure and obstacle avoidance, showcasing a significant improvement in foraging success and scalability over traditional foraging algorithms. In future work, we will leverage Federated Learning (FL) to build an efficient, distributed, scalable, and secure robot swarm tailored for foraging tasks.

I. INTRODUCTION

Swarm robotics leverages groups of autonomous robots to perform complex tasks collaboratively, with foraging being a fundamental application. Foraging involves searching, retrieving, and transporting resources from an undetermined, dynamic environment back to a central location, also known as a central collection zone [11]. This task is critical and applicable in domains such as search and rescue, environmental monitoring, planetary exploration, and disaster response, where adaptability and efficiency are paramount. The ability of swarms to operate with distributed decision-making, scalability, and fault tolerance makes them particularly well-suited for foraging [12].

Traditional algorithms, such as Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO), have been instrumental in enabling collective behaviors like exploration and resource retrieval [7], [3], [13]. However, their fixed rules and parameter settings often limit their ability to adapt to varying resource distributions, obstacles, and environmental changes [14]. Similarly, the Central Place Foraging Algorithm (CPFA) [3], which employs pheromone-based communication and memory to optimize resource collection paths, and the Distributed Deterministic Spiral Algorithm (DDSA) [9], which uses systematic spiral patterns to ensure exhaustive searches. In [2], it offers reliable solutions to scale

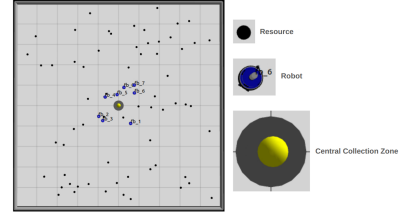


Fig. 1. An illustration of ARGoS robot simulation environment. The arena size is 10×10 m. 8 robots are deployed around the center, and 64 resources are distributed randomly.

effectively to larger swarms or dynamic conditions. In [1], it shows the AI-trained models can evolve adaptive foraging behaviors.

This study introduces distributed learning and the NEAT [20] to optimize swarm foraging behaviors in search patterns, pathfinding in delivering resources, and avoiding obstacles. NEAT evolves both the topology and weights of neural networks, enabling robots to dynamically adapt their strategies to diverse environmental conditions. Integrating a penalty-reward system inspired by reinforcement learning prompts efficient resource retrieval, minimizes redundant exploration, and penalizes actions that waste energy or reduce collaboration. Swarm robotics focuses on designing decentralized algorithms that enable simple robots to perform tasks like foraging collectively. A key challenge is ensuring these strategies remain effective under real-world noise and uncertainty. [15] compare a deterministic and a bio-inspired stochastic foraging algorithm, highlighting how real-world testing can shift conclusions drawn from simulations. Recent studies have further highlighted the importance of emergent communication in enhancing foraging efficiency. The work in [17] has shown that evolved swarms controlled by Spiking Neural Networks can develop communication strategies that significantly improve collective foraging performance. Additionally, by grounding decision-making in sensorimotor interactions with the environment, our system reflects the role of embodiment in shaping adaptive behavior, as discussed in [18].

By comparing the proposed approach to algorithms like CPFA and DDSA, this research demonstrates that it improves cumulative reward fitness and addresses key challenges like congestion around the collection zone and resource allocation inefficiencies. This integrated method fosters emergent behaviors that adapt dynamically to environmental changes, offering a scalable, robust solution that outperforms traditional algorithms in foraging tasks.

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II. RELATED WORK

The design of efficient and adaptive swarm robotic controllers has been a persistent challenge in robotics research, particularly for tasks such as foraging, exploration, and resource collection. Traditional approaches like CPFA and DDSA have been widely used to address swarm foraging [8]. The CPFA combines hand-coded behaviors with adaptively tuned parameters to optimize resource collection while leveraging pheromone-based communication to guide swarm members. Similarly, DDSA employs pre-defined geometric search patterns, ensuring exhaustive coverage of the environment. However, these approaches are often constrained by fixed parameterization and limited scalability in dynamic environments, particularly with larger swarm sizes [3] [4].

The Distributed Deterministic Spiral Algorithm (DDSA) has proven effective for swarm foraging but struggles with scalability due to robot collisions. To address this, the Multiple-Distributed Bidirectional Spiral Path Planning Algorithm proposed which improves coordination and reduces interference in larger swarms [10]. Recent advancements in evolutionary robotics have demonstrated the potential of genetic algorithms to evolve robotic controllers. For instance, [5] introduced a neural-endocrine system with a feed-forward backpropagation neural network to develop swarm behaviors for foraging. However, this approach required significant manual intervention to retrain or reprogram the network when new behaviors were needed. More recently, [6] explored evolving parameters for task-specialized heterogeneous swarms, which allowed robots to specialize in roles such as collectors and droppers. While effective, these methods focused primarily on fixed behaviors or heterogeneity, limiting their adaptability across diverse conditions.

The NEAT algorithm, introduced by Stanley and Miikkulainen, addresses these limitations by evolving both the topology and weights of neural networks [20]. NEAT has shown promise in robotic applications, as it enables the discovery of compact and efficient networks through incremental complexity growth. The work of Ericksen et al. applied NEAT to evolve controllers for swarm foraging in various resource distributions, demonstrating its capacity to produce scalable and generalizable behaviors. This study revealed that NEAT-based controllers, like the proposed NEAT-FA algorithm, outperform traditional hand-coded algorithms like CPFA and DDSA, especially in handling larger swarm sizes without congestion at the central collection zone.

Building on these foundations, our research employs NEAT with a penalty-reward system to further optimize swarm foraging. We evaluate our approach across randomly distributed environments, emphasizing scalability and adaptability. By incorporating reward mechanisms for efficient resource retrieval and penalties for redundant or energy-inefficient actions, we align our method with reinforcement learning principles to encourage emergent collaborative behaviors. Additionally, our experimental setup directly compares the performance of our NEAT-evolved controllers to

CPFA, enabling a clear analysis of cost-efficiency, scalability, and adaptability.

In summary, this study contributes to the growing body of research on automated controller evolution by integrating NEAT for optimizing the foraging behavior and obstacle avoidance, advancing beyond traditional algorithms and even existing NEAT implementations in swarm foraging tasks.

III. PROPOSED METHODS

Despite advancements in swarm robotics, challenges persist:

- 1) **Adaptability:** Fixed rules limit traditional algorithms' ability to respond dynamically to environmental changes, such as varying resource distributions, unpredictable obstacles, or shifts in terrain. This lack of flexibility results in inefficiencies in resource retrieval and navigation.
- 2) **Scalability:** As swarm sizes increase, larger groups of robots often experience congestion at collection zones, inefficient resource allocation, and communication bottlenecks. These issues reduce overall performance and make traditional algorithms unsuitable for large-scale deployments.
- 3) **Energy Efficiency:** Inefficient exploration strategies lead to redundant movements and higher energy consumption during resource retrieval tasks. This significantly impacts the operational lifespan of robots, particularly in resource-constrained environments.
- 4) **Coordination Complexity:** Coordinating multiple robots with limited sensing and computational capabilities is inherently challenging. Traditional algorithms struggle to achieve optimal division of labor or collaborative behaviors in dynamic environments.

To address the challenges in swarm robotics foraging tasks, we propose the NEAT-Foraging Algorithm (NEAT-FA), which integrates dynamic neural network evolution with a penalty-reward mechanism to optimize swarm behavior in dynamic environments with obstacles. The key components of the proposed idea are as follows:

A. *NeuroEvolution of Augmented Topologies (NEAT)*

- **Search Optimization:** Efficiently locating resources in random, semi-clustered, and clustered distributions.
- **Obstacle Avoidance:** Navigating around barriers without compromising task efficiency.
- **Resource Delivery:** Transporting collected resources to the central collection zone through optimized paths.

B. *Penalty-Reward Mechanism*

- **Rewards:** +1 for resource retrieval and +2 for successful delivery to the central collection zone.
- **Penalties:** Fuel costs applied every 5 seconds with values (-0.01, -0.001, -0.005, -0.0001), penalizing unnecessary movements or idle time.

$$Reward = \begin{cases} +1, & \text{for picking up a resource} \\ +2, & \text{for delivering the picked up resource} \\ -c, & \text{for taking extra time} \end{cases}$$

$$R = R_{pickup} + 2R_{delivery} - CE_{consumed} \quad (1)$$

, where R is resources, C is cost factor, and E is energy consumed.

C. Optimization Objectives

- **Adaptability:** Real-time adjustment to dynamic resource distributions and obstacle layouts.
- **Energy Efficiency:** Minimizing unnecessary movements to reduce energy consumption.
- **Scalability:** Ensuring efficient performance for larger swarm sizes without congestion or communication bottlenecks.

D. Comparative Analysis

- Unlike traditional pheromone-based coordination in CPFA, our penalty-reward system enables emergent collaboration through fitness-driven neural evolution, eliminating centralized communication bottlenecks.
- Benchmarked against CPFA and NEAT-FA without penalty in:
 - Cumulative reward fitness.
 - Task completion rates.

IV. DESIGN

To further enhance the system's adaptability, efficiency, and scalability, we have extended its application to environments with obstacles and directly compared its performance to the CPFA.

We conducted a series of simulations in a swarm robotics environment involving the central point foraging task. The experimental setup was designed to assess the adaptability of NEAT-FA in navigating obstacle-rich environments, its efficiency in optimizing resource retrieval paths, and its scalability across different swarm sizes. Our comparison included baseline NEAT-FA (without penalties) and CPFA to evaluate how penalty-based NEAT-FA improves performance in terms of cumulative rewards, energy efficiency, and trajectory optimization.

The experiments were conducted using the ARGOS multi-robot simulator. The arena was modeled as a bounded two-dimensional space measuring 10 meters by 10 meters. Obstacles were randomly placed within the environment to simulate real-world complexities. The arena was populated with resources distributed under three scenarios: random, semi-clustered, and clustered distributions. These scenarios were chosen to test the adaptability of NEAT-FA across varying levels of environmental complexity.

The robots used in these experiments were configured with basic sensory capabilities for detecting resources and avoiding obstacles. They relied on pheromone-based communication for collaboration and operated under limited computational power to reflect real-world hardware constraints. The

robots used a simple wheeled locomotion system, ensuring that the experimental setup was consistent with physical swarm robotics platforms.

NEAT was configured to evolve neural network topologies and weights across generations dynamically. The parameters for NEAT were set as follows: a population size of 100 neural networks per generation, a mutation rate of 0.9, a crossover rate of 0.3, and a generational limit of 100. These settings allowed NEAT to evolve efficient controllers for resource retrieval, obstacle avoidance, and energy-efficient navigation.

The fitness function used in this study optimized searching, delivering resources, and avoiding obstacles while minimizing redundant movements. Robots were rewarded with +1 for successfully picking up resources and +2 for delivering resources to the central collection zone. A penalty (living cost) was applied every 5 seconds with values (-0.01, -0.001, -0.005, -0.0001) to discourage unnecessary movements or idle behavior.

To evaluate performance, we compared penalty-based NEAT-FA against baseline NEAT-FA and CPFA across all resource distribution scenarios. Metrics such as cumulative rewards, energy efficiency, trajectory patterns, and obstacle navigation success rates were recorded during simulations. This comparative analysis highlighted how penalty-based NEAT-FA not only outperformed CPFA but also addressed key limitations of baseline NEAT-FA by reducing redundant exploration and improving task completion rates.

By incorporating obstacle-avoidance optimization into NEAT's dynamic evolution framework and comparing it against CPFA in obstacle-rich environments, we demonstrated that penalty-based NEAT-FA offers significant improvements in adaptability, scalability, and energy efficiency for swarm robotics applications.

V. EXPERIMENTAL SETUP

We have implemented the NEAT for optimizing the search and delivering resources. We will consider the optimization in the environment with obstacles. We will also leverage federated learning for NEAT. We conducted a series of simulations in a swarm robotics environment involving the central point foraging task. Our code and videos are available at Github¹ and Youtube².

The experimental setup was designed to assess the adaptability, efficiency, and scalability of NeatFA compared to baseline NeatFA without considering optimizing the shortest paths in searching and delivering resources. The following components define the experimental framework:

Simulation Environment: The experiments were conducted using the ARGOS multi-robot simulator. The arena was modeled as a bounded 2-dimensional space, where X and Y represent the width and height of the area, respectively 10 meters each. The experiment populates the arena during three

¹<https://github.com/MARSLab-UTRGV/neatfa-tameem/tree/Pigar-Semi-Cluster-Training>

²<https://www.youtube.com/playlist?list=PLm1FtL2xiqsD3tnQK5FnNYhGtV4cUPXge>

scenarios: random, semi-clustered, and clustered distributed resources.

Robot Configuration: The experiment involved swarm sizes of 8 and 16, including foraging robots, each equipped with basic sensory capabilities for obstacle avoidance, resource detection, and pheromone communication. The robots used a simple wheeled locomotion system and operated under limited computational power, reflecting the constraints of physical swarm robotics platforms.

NEAT Configuration: NEAT evolves neural network topologies and weights across generations to optimize resource retrieval efficiency. The foraging controllers were initialized using the NEAT algorithm. The parameters were set as follows: Population Size: 100; Mutation Size: 0.9; Crossover Rate: 0.3; Generational Limit: 100.

Fitness Function: The fitness function was designed to optimize the searching, delivering, and obstacle avoidance. The reward is 1 for picking up resources and 2 for delivering the resources to the central collection zone. We defined a penalty (living cost) with 5 different values (-0.01, -0.001, -0.005, and -0.0001) every 5 seconds.

VI. RESULTS

The performance of the proposed NEAT-Foraging Algorithm (NEAT-FA) was evaluated through a series of simulations designed to compare its effectiveness against baseline NEAT-FA (without penalties) and the Central Place Foraging Algorithm (CPFA). The results demonstrate that penalty-based NEAT-FA significantly outperforms both baseline NEAT-FA and CPFA in terms of adaptability, energy efficiency, and resource retrieval rates across all tested scenarios.

A. Cumulative Rewards

The cumulative rewards, which measure the total efficiency of resource retrieval and delivery, were significantly higher for penalty-based NEAT-FA compared to the other algorithms. In random resource distributions, cost factors of 0.001 and 0.0001 yielded the best performance, with robots demonstrating more direct trajectories and efficient collaboration. For semi-clustered and clustered distributions, the differences in performance between cost factors were less pronounced due to the structured nature of resource layouts, but penalty-based NEAT-FA still outperformed baseline NEAT-FA and CPFA. As shown in Figures 2, 3, and 4, we trained 16 robots in 3 different distributions and with 4 different cost factors. This experiment identified the best foraging performance in training.

B. Energy Efficiency

Penalty-based NEAT-FA demonstrated improved energy efficiency by minimizing redundant movements. Fig. 5 shows that baseline NEAT-FA produced circular or inefficient trajectories, leading to increased energy consumption. In contrast, penalty-based NEAT-FA resulted in straight and purposeful trajectories from resource depots to the central collection zone, reducing travel time and energy expenditure.

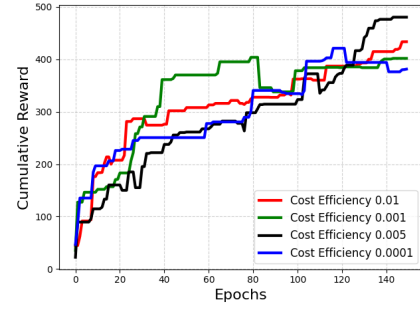


Fig. 2. 16 robots in randomly distributed resources.

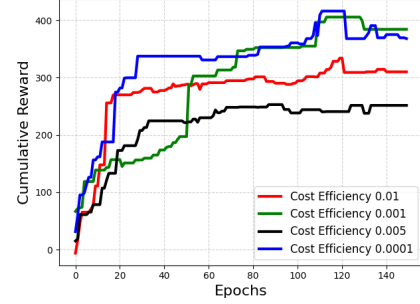


Fig. 3. 16 robots in semi-clustered distributed resources.

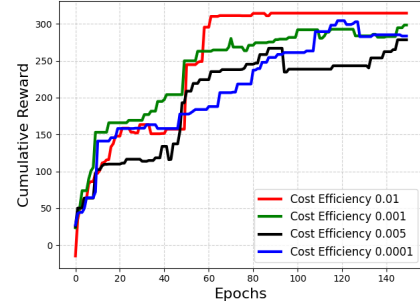


Fig. 4. 16 robots in clustered distributed resources.

This improvement was particularly evident in random distributions, where robots needed to explore more extensively before locating resources.

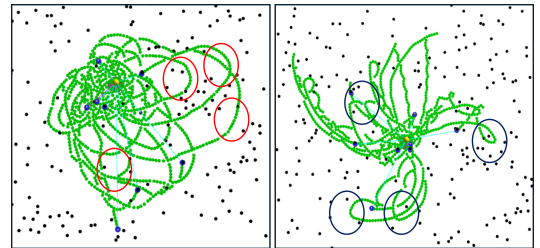


Fig. 5. 8 robots run in the same environment with the same time frame, shows how existing NeatFA generated a long circular path (left) compared to our version of NeatFA with a penalty.

C. Obstacle Navigation

In environments with obstacles, penalty-based NEAT-FA exhibited superior obstacle avoidance compared to CPFA and

baseline NEAT-FA. The evolved neural networks enabled robots to dynamically adapt their paths around obstacles without compromising task efficiency. This adaptability was critical in maintaining high cumulative rewards and energy efficiency in obstacle-rich scenarios.

Figures 6, 7, and 8 demonstrate that NEAT-FA outperforms CPFA in total resource collection across increasing swarm sizes in obstacle-rich environments. This consistent trend supports the conclusion that NEAT-FA maintains high task performance and scales effectively even when environmental complexity and agent density increase.

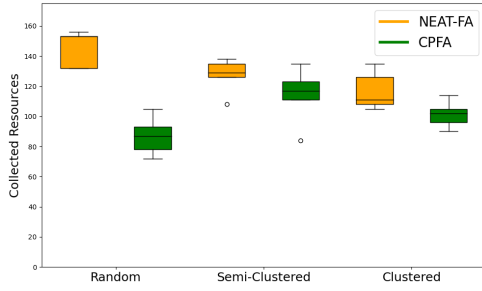


Fig. 6. The collected resources with 4 robots and obstacles.

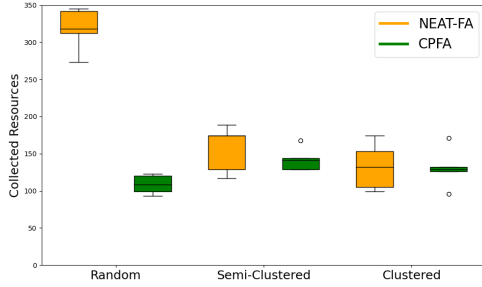


Fig. 7. The collected resources with 8 robots and obstacles.

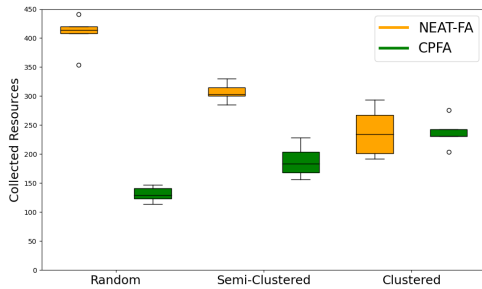


Fig. 8. The collected resources with 16 robots and obstacles.

D. Scalability

The experiments tested swarm sizes of 8 and 16 robots to evaluate scalability. Penalty-based NEAT-FA maintained consistent performance across different swarm sizes, avoiding congestion at collection zones and ensuring efficient communication among robots. CPFA, on the other hand, struggled with scalability due to its reliance on pheromone-based communication, which became less effective as swarm size increased.

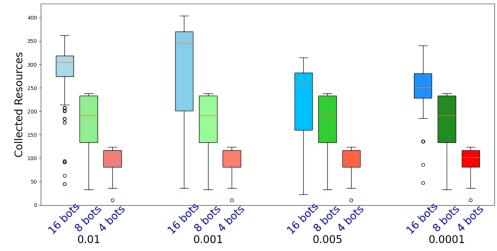


Fig. 9. The collected resources in the random distribution.

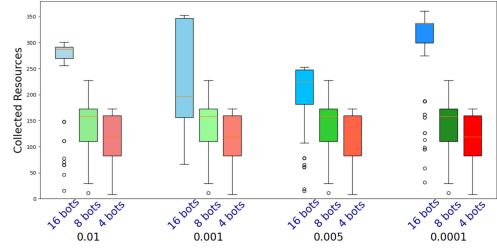


Fig. 10. The collected resources in semi-clustered distribution.

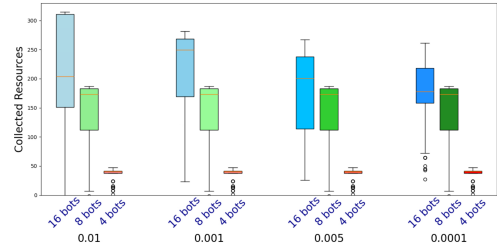


Fig. 11. The collected resources with 4, 8, and 16 robots in the clustered distribution.

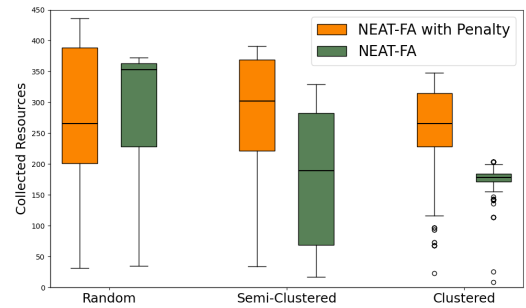


Fig. 12. Performance Comparison Between Existing NeatFA vs Penalty-Based NeatFA Algorithm

Figures 9, 10, and 11 show that 0.001 is the most suitable cost factor, as it performed well in 3 different distributions. At the end, we compare the previous NEAT-FA algorithm with our penalty-based NeatFA algorithm. As we can see from Fig. 12, our algorithm performed better in all 3 distributions. The result can be seen in the simulation Fig. 5, where the robots are generating fewer long curves compared to the previous one. This helps the robots to collect more resources in less time.

VII. CONCLUSIONS AND FUTURE WORK

This research investigates the effectiveness of the NEAT framework in further optimizing robot swarms when considering the obstacle avoidance and living costs of foraging tasks. We have shown that the existing NeatFA generates unnecessary circular motions, while our proposed NeatFA reduces this kind of circular motion and outperforms the existing NeatFA by collecting more resources in a shorter time. By considering the living costs, the system reduces redundant exploration, enhances resource retrieval, and minimizes travel time, outperforming the NeatFA and traditional CPFA algorithms. We will improve the scalability and privacy of the training framework by combining federated learning. The findings underscore the potential of NEAT-based approaches for adaptive swarm behaviors, with promising applications in fields such as search and rescue and environmental monitoring. Future work will focus on optimizing the environment with obstacles and scalability and privacy by distributed federated learning [19]. We'll also implement our new reward function to see its performance in the simulation. Below is the function description:

$$F = \frac{\log(1 + R_{pickup}) * \log(1 + R_{delivery})}{1 + E_{consumed}/E_{total}} \quad (2)$$

With this equation 2, if the robot uses too much energy, then it will be given less reward. This way, we can encourage the robot to deliver the resource with less amount of energy.

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